Sensorless Commutation Deviation Correction of Brushless DC Motor With Three-Phase Asymmetric Back-EMF

Gang Liu, Xi Chen, Xinxiu Zhou, *Member, IEEE*, and Shiqiang Zheng, *Member, IEEE*

***Abstract*—In view of the sensorless commutation deviation of brushless dc motor with asymmetric three-phase back electromotive force (EMF), this article presents a correction method of commutation deviation based on the principle of current consistency before and after commutation to achieve optimal commutation. First, this article analyzes the different factors resulting in the commutation deviation of the motor with asymmetric three-phase back-EMF, and then points out the deviation disadvantage. Based on the analysis, a commutation deviation correction method based on current consistency in ideal optimal commutation is proposed. At the same time, in order to eliminate the influence of waveform distortion caused by winding inductance in the commutation process, dc-link current reconstructor is proposed. Based on the reconstructed current value after commutation, an optimal commutation phase estimator based on commutation current consistency is designed to estimate the position of each commutation point. Finally, experimental results confirm that the proposed method can achieve high-precision deviation correction and optimal commutation at each commutation point in the case of asymmetric three-phase back-EMF.**

***Index Terms*—Asymmetric back electromotive force (EMF), brushless dc (BLDC) motor, commutation deviation correction, optimal commutation, sensorless control.**

I. Introduction

BRUSHLESS dc motor (BLDCM) has been widely used in household appliances, industry, and aerospace fields, such as molecular pumps, compressors, blowers, attitude control flywheels, gyroscopes, etc., due to its advantages of high energy density, high torque-current ratio, and small volume [1]. Especially for space applications, which require high energy efficiency and excellent performance, BLDCM is very suitable. For example, in a magnetically suspended control moment gyroscope (MSCMG) that is considered to be an advanced attitude control actuator of spacecraft, BLDCM is used as an inertial wheel driver to provide high stable and large angular momentum. The control performance of BLDCM has a significant impact on the performance of the attitude control actuator [2]. One of the most important signals required for BLDCM control is the rotor position. In the traditional BLDCM drive system, six discrete rotor position signals are needed, which are typically detected by three Hall sensors mounted on the motor stator. However, the application of mechanical sensors brings potential risks to the reliability of motor operation, which severely limits the application of BLDCM in harsh environments such as space. Therefore, sensorless control is critical for BLDCM drives, especially for special applications. Several sensorless control methods for BLDCM have been widely studied in the past decades. The rotor position of the BLDCM can be detected by the zero-crossing point (ZCP) of back electromotive force (back-EMF) [3], [4], the third harmonic of back-EMF [5], [6], or other characteristic signals.

Traditional BLDCM control is based on the basic assumption that the three-phase back-EMF is completely symmetrical whether using position sensor or sensorless technical. However, due to the limitation of the design and manufacturing process, it is difficult to guarantee the assumption that the three-phase back-EMF is completely symmetrical, especially when the magnetic gap between the rotor and stator of the motor fluctuates, which is manifested in the fact that the amplitude and bias of the three-phase back-EMF are not exactly the same, as shown in Fig. 1. When the three-phase back-EMF are no longer completely symmetrical, Hall or sensorless detection methods will inevitably produce position detection deviations, which will cause severe current and torque ripple. In addition, this deviation is very similar to the position detection deviation introduced by the nonbalanced mounted Hall sensors described in some research works [7]–[10]. Many researchers have proposed a variety of compensationmethods for position detection deviations caused by unbalanced mounted Hall sensors. Beccue *et al.* compensated the commutation deviation introduced by the unbalanced Hall sensors and reduced the torque ripple of the motor [7]. Samoylenko *et al.* compared several extrapolation and filtering algorithms to deal with unbalanced commutation signals and,

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The authors are with the School of Instrumentation Science and Optoelectronics Engineering, Beihang University (BUAA), Beijing 100191, China, and also with the Ningbo Institute of Technology, Beihang University (BUAA), Ningbo 315800, China (e-mail: lgang@buaa.edu.cn; chen_xi@buaa.edu.cn; 09757@buaa.edu.cn; zhengshiqiang@buaa.edu.cn).

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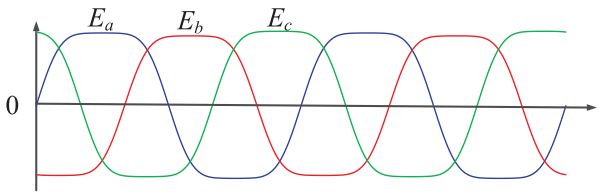


Fig. 1. Three-phase asymmetric back-EMF of BLDCM.

thus, effectively improved the dynamic performance, but its signal processing algorithm is relatively complex [8]. In addition to the trapezoidal back-EMF motor, Park *et al.* proposed an online adjustment method for the commutation deviation of sinusoidal wave back-EMF BLDCM caused by the installation error of Hall sensors. Compared with the open-loop compensation method, this closed-loop method improves the correction accuracy of commutation deviation [9], [10].

In order to avoid misdetection caused by modulation noise and other interference signals, a low-pass filter (LPF) with quite a low cut-off frequency is often employed to process the detection signal when the sensorless method is used. Although the use of LPF eliminates misdetection, it also introduces a large signal delay, resulting in commutation deviation. Moreover, different kinds of nonideal factors in the motor control system will also bring deviation to the commutation of the motor. Shen and Tseng introduced the analysis and calculation of the detection error and motor commutation angle, then gave an error compensation circuit to ensure correct commutation [11]. Jiang *et al.* proposed a digital filter processing program without phase delay to replace the LPF in order to obtain accurate back-EMF ZCPs [12]. Furthermore, some closed-loop correction methods have also been extensively studied. Based on the symmetry of the voltage at the beginning and end of commutation interval, Jang and Kim used a proportional integral (PI) regulator to correct the commutation deviation [13]. Wu *et al.* pointed out that the freewheeling current error can accurately reflect the commutation deviation and propose a new method to correct the commutation deviation [14]. Wu *et al.* used coordinate transformation to compensate for rotor position detection errors by controlling the angle between two coordinates in real time [15].

Recently, a number of novel and more precise commutation deviation correction methods have been studied and applied. Li *et al.* proposed a two-stage commutation error compensation method based on the detailed error analysis of traditional line-to-line voltage ZCPs detection to achieve high-precision commutation [16]. Lee and Kong proposed an algorithm for determining the phase lead angle from the Fourier series of phase voltage waveforms and designed a logic circuit to implement the algorithm [17]. Gu *et al.* analyzed the commutation deviation caused by different factors, and pointed out that the commutation deviation was equal to the internal power factor angle, and then proposed a phase-locked loop (PLL) based correction method for commutation deviation [18]. Zhou *et al.* used the symmetry of phase voltage to calculate the voltage difference and proposed a commutation deviation correction method based on a neural network fuzzy controller [19]. Based on the analysis of the relationship between commutation deviation and dc-link current, Li *et al.* proposed a deviation correction strategy based on the model-free adaptive control method [20]. Although aforementioned methods can effectively correct or compensate the commutation deviation caused by different factors, they are all based on the symmetrical assumption, especially that the voltage or current are symmetrical before and after the conduction interval. However, the symmetry assumption is no longer suitable to three-phase asymmetric back-EMF motors. Park *et al.* compensated for the ZCP detection bias caused by dc voltage drop, offset, and asymmetric back-EMF to obtain a balanced commutation point, but the asymmetric back-EMF motor was commutated with equidistant balanced commutation points [21]. The suppressing effect of current ripple for asymmetric back-EMF motor commutated with balanced commutation points needs further verification. For the high-precision commutation of three-phase asymmetric back-EMF motors, as far as known, no relevant research works have been reported. Therefore, it is worthwhile to study the sensorless high-precision commutation control method considering the asymmetric back-EMF.

This article first analyzes various factors affecting the commutation deviation of three-phase asymmetric back-EMF motors and proposes a commutation deviation correction method based on commutation current consistency. The method includes a current reconstructor (CR) designed to eliminate the waveform distortion caused by winding inductance during commutation, an optimal commutation phase estimator (OCPE) based on commutation current consistency, and a correction phase shifter. The final experiment results show that the proposed method can achieve high-precision optimal commutation by correcting the commutation deviation of each commutation point under the condition of three-phase asymmetric back-EMF.

II. Sensorless Commutation Deviation of Three-Phase Asymmetric Back-EMF Motor

A. BLDCM Mathematical Model

The BLDCM used in the high-speed rotor system of MSCMG is generally designed as coreless cup-type stator in order to reduce eddy current loss. At the same time, the inductance of the motor winding is very small (less than 50 μH), which results in more high-frequency noise interference in the coil current when pulsewidth modulation (PWM) is applied. The high-frequency interference can be effectively alleviated by the PAM driving mode, which is using a buck converter on the dc side, as shown in Fig. 2 [22]. Among them, the buck converter is responsible for dc voltage regulation through PWM modulation, while the three-phase full-bridge inverter works in the non-PWM mode only for motor commutation every 60 electric degrees. Under the assumption that the three-phase resistance and inductance are constant and equal, and neglecting the saturation of the motor core, the phase voltage equation of BLDCM is written as

$$U_{xg} = L\frac{di_x}{dt} + Ri_x + E_x + U_{ng} \tag{1}$$

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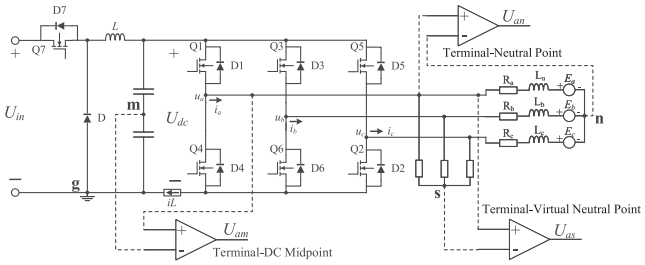


Fig. 2. BLDCM driver configuration with buck converter and schematic diagram of several sensorless ZCPs detection methods.

where U_{xy} denotes the voltage between x and y, L denotes the phase self-inductance, R is resistance, E_x and i_x denote the back-EMF and current of phase x, $x = a, b, c$. Note that, the stator of the surface-mounted motor used in this article has no iron core, the magnetic field coupling effect between three-phase windings is very weak, and the mutual inductance value is almost unmeasurable, so the mutual inductance has been neglected in the following calculation and analysis.

The back-EMF harmonic model considering three-phase asymmetry can be expressed as

$$\begin{aligned} cE_x &= \omega_e k_e \left[\sum_{m=1}^{\infty} A_{xm} \sin\left(m\left(\theta_e + n\frac{2\pi}{3} \right) \right) + D_x \right] \\ &= \omega_e k_e e_x\left(\theta_e\right) = \omega_e k_e e_x\left(\omega_e t\right) \end{aligned} \tag{2}$$

where ω_e represents the electrical speed, k_e represents the back-EMF coefficient, θ_e represents the electrical angle position, A_{xm} represents the mth harmonic coefficient of the x phase back-EMF, n is $0, -1, 1$ when x is a, b, c, D_x represents the dc offset of the x phase back-EMF, and e_x represents the x phase back-EMF waveform function. The corresponding coefficients can be obtained by offline measurement.

B. Commutation Deviation Caused by Three-Phase Asymmetric Back-EMF Sensorless ZCPs Detection

Commonly used methods for detecting ZCPs of back-EMF include terminal-neutral point voltage detection, terminal-virtual neutral point voltage detection, and terminal-dc side midpoint voltage detection. The ZCPs obtained by these methods need to be delayed by 30 electrical angles before they can be used as the real commutation points of the motor. Fig. 2 illustrates schematic diagram of three different ZCP detection methods.

1) Terminal-Neutral Point Voltage Detection Method: In the terminal-neutral point voltage ZCP detection method, the neutral voltage signal needs to be extracted from the stator winding by the conductor. This method compares the terminal voltage with the neutral point voltage to obtain ZCPs. When the detection phase is nonconducting, the detected voltage is exactly the real back-EMF as

$$\begin{cases} U_{an} = E_a \\ U_{bn} = E_b \\ U_{cn} = E_c \end{cases}. \tag{3}$$

2) Terminal-Virtual Neutral Point Voltage Detection Method: In the terminal-virtual neutral point voltage ZCP detection method, Y-connected resistor network is used to provide virtual neutral voltage signal. This method compares the terminal voltage with the virtual neutral point voltage to obtain the ZCPs.

Taking phase A as nonconduction, for example, the following relations can be obtained according to the constraints of Y-type connections and Kirchhoff's law:

$$\begin{aligned} U_{ag} &= U_{as} + U_{sg} = U_{an} + U_{ng} \\ U_{sg} &= \frac{1}{3}\left(U_{ag} + U_{bg} + U_{cg}\right) \\ U_{ng} &= \frac{1}{3}\left(U_{ag} + U_{bg} + U_{cg} - E_a - E_b - E_c\right). \end{aligned} \tag{4}$$

Then, the following equation can be known as

$$U_{as} = U_{ag} - U_{sg} = \frac{1}{3}\left(2E_a - E_b - E_c\right). \tag{5}$$

Similarly, for B and C, when they are nonconducting phases, there are

$$\begin{aligned} U_{bs} &= U_{bg} - U_{sg} = \frac{1}{3}\left(2E_b - E_c - E_a\right) \\ U_{cs} &= U_{cg} - U_{sg} = \frac{1}{3}\left(2E_c - E_a - E_b\right). \end{aligned} \tag{6}$$

3) Terminal-DC Side Midpoint Voltage Detection Method: In the terminal-dc midpoint voltage ZCP detection method, two identical dividing resistors provide dc midpoint voltage. The voltage comparator compares the terminal voltage with the midpoint voltage.

Take the phase A as nonconduction, for example. At this time, the following equation can be obtained:

$$U_{am} = \frac{1}{2}\left(2E_a - E_b - E_c\right). \tag{7}$$

Similarly, for B and C, when they are nonconducting phases, there are

$$\begin{aligned} U_{bm} &= \frac{1}{2}\left(2E_b - E_c - E_a\right) \\ U_{cm} &= \frac{1}{2}\left(2E_c - E_a - E_b\right). \end{aligned} \tag{8}$$

The measured voltages and corresponding ZCPs distribution by three methods are shown in Fig. 3. Although the voltages measured by the various methods are different, all the ZCPs are affected by the asymmetric back-EMF and there are slight phase deviations between each other.

In theory, the so-called optimal commutation points should be the intersection points of two adjacent back-EMF, that is, the points where $E_a = E_c$, $E_a = E_b$, $E_b = E_c$, which are 30° lagged by ZCPs in an ideal symmetric back-EMF motor. Define the phase of the six optimal commutation points deviating from the detected ZCPs as φ_{z1-6}.

C. Commutation Deviation Caused by Other Factors

Besides the effect of asymmetric three-phase back-EMF on the phase deviation of the ZCP detection method, various signal

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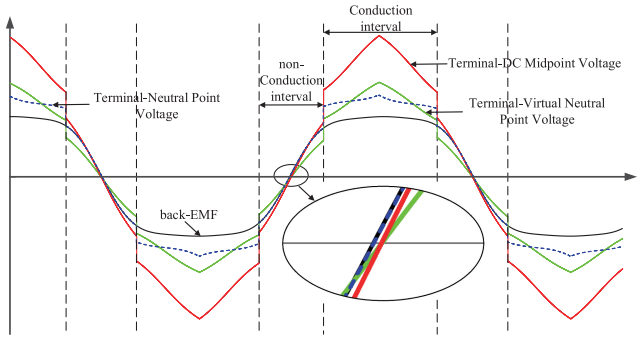


Fig. 3. Schematic diagram of measured voltage and actual ZCPs by several sensorless ZCPs detection methods.

processing circuits, digital processors, and software also introduce phase deviation, of which the phase delay introduced by LPF is the majority part. The cut-off frequency of LPF designed to eliminate noise interference is very low, so large phase delay is unavoidable. The LPF is generally approximated to a first-order inertia unit as

$$\mathrm{LPF}(s) = \frac{\omega_c}{s + \omega_c} \tag{9}$$

where w_c is the cut-off frequency, w_e is the electrical speed frequency. Then, the phase delay can be obtained by the phase-frequency characteristics as

$$\varphi_{\mathrm{LPF}} = -\arctan\left(\omega_e/\omega_c\right). \tag{10}$$

In addition to the aforementioned large phase delay caused by LPF, there are some phase deviations that are difficult to determine and measure caused by other factors, such as AD sampling, software calculation, which are defined as φ_{other}.

D. Total Commutation Deviation

According to the analysis of commutation deviation of the three-phase asymmetric back-EMF motor, the total deviation caused by various factors can be summarized as

$$\varphi = \varphi_{z1-6} + \varphi_{\mathrm{LPF}} + \varphi_{\mathrm{other}}. \tag{11}$$

Asymmetric back-EMF related deviation is usually small (within 5°) and does not vary with the speed, while deviation caused by the LPF delay will gradually increase with the increase of the rotational speed (up to a dozen degrees). Two aliasing deviations can result in large total commutation deviations at medium to high speeds.

III. Influence of Commutation Deviation and Correction Principle

A. Current Waveform Distortion

When the phase deviation exists in the commutation of the motor, the phase current and the dc-link current will be significantly distorted, which is mainly caused by the back-EMF in the nonideal conduction interval. Fig. 4 shows the waveforms of phase current and dc-link current of the asymmetric back-EMF motor in the advanced 10° commutation, normal commutation, and lagged 10° commutation, respectively. Obviously, as can be seen, when commutation is advanced, the phase current at front end is higher than that at back end in the 60° conduction interval. When motor commutate normally, the phase current at front end and back end are basically the same. However, when commutation is lagged, the phase current at back end is higher than that at front end in the 60° conduction interval. In addition, due to the influence of three-phase asymmetric back-EMF, the current waveforms in each 60° conduction interval are no longer symmetrical but slightly different.

Waveform distortion will increase peak current, increase torque ripple, reduce motor efficiency, and increase motor power consumption. Therefore, it is necessary and urgent to correct the commutation deviation of the motor with asymmetric back-EMF.

B. Correction Principle

According to the waveform analysis of phase current and dc-link current, it can be seen that the commutation deviations can be corrected by using the symmetrical characteristics of current waveforms at the front and back ends of each 60° conduction interval, which is also the basic principle of the traditional symmetry-based correction method. However, for motors with asymmetric back-EMF, the symmetry-based correction method cannot achieve high-precision commutation. Although the detailed parameters of the back-EMF can be obtained by offline sampling in advance, and the relationship between ZCPs and the optimal commutation points can be calculated and compensated in the offline compensation method. This ideal method does not consider the phase delay caused by other factors such as LPF. In the traditional method, the effects of asymmetric back-EMF and phase delay need to be compensated separately, which greatly increases the system complexity.

The consistency of current before and after commutation can also be used as an indicator of commutation deviation, regardless of whether the motor's back-EMF is symmetrical or not. Fig. 5(a) shows the current waveform of the symmetrical back-EMF motor without considering the influence of winding inductance. A and B are the points before and after the commutation instant and there is no phase separation between them and the commutation point. It can be seen that point A representing the current before commutation is equal to point B representing the current after commutation. If the back-EMF of the motor is asymmetrical, the current waveforms in each conduction interval are slightly different, but the current values before and after commutation are consistent without the influence of winding inductance, as shown in Fig. 5(b). Since no freewheeling is considered, these two points are sufficient to reflect the consistency of the current before and after the commutation regardless of the asymmetry in back-EMF. However, in reality, the influence of winding inductance on current cannot be ignored. In order to avoid the interference of freewheeling current at the current sampling point, many researchers deviate the current sampling points a same specific phase angle before and after the commutation point to escape freewheeling region, as shown in Fig. 5(c). The current values at shifted current sampling points A and

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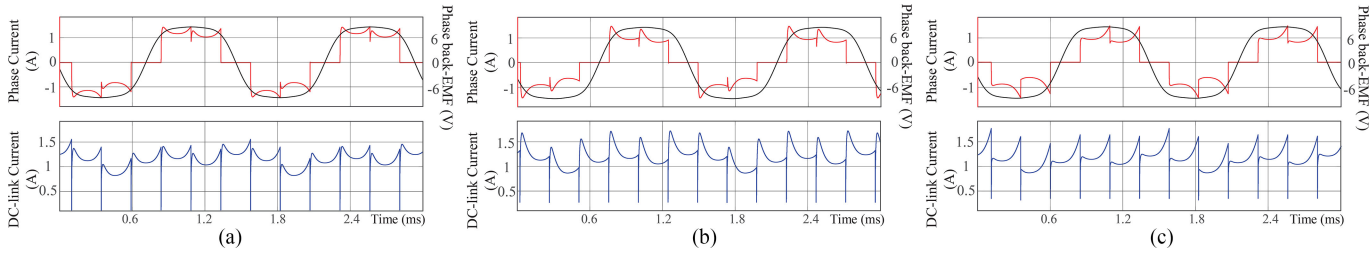


Fig. 4. Back-EMF, phase current, and dc-link current waveform of BLDCM. (a) With optimal commutation. (b) With advanced commutation. (c) With lagged commutation.

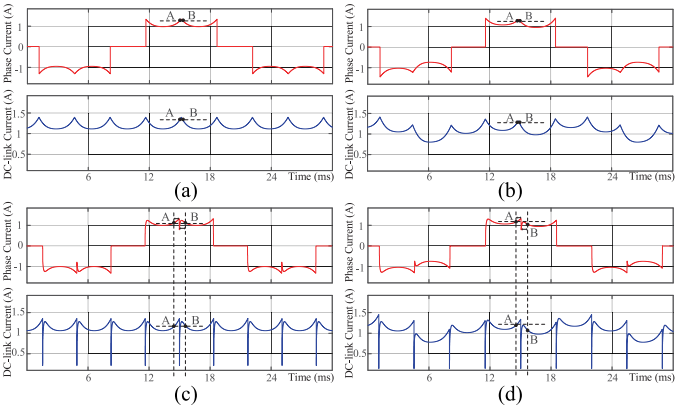


Fig. 5. Comparison of correction methods based on current consistency. (a) Symmetric back-EMF without freewheeling. (b) Asymmetric back-EMF without freewheeling. (c) Symmetric back-EMF with freewheeling. (d) Asymmetric back-EMF with freewheeling.

B still keep the consistency due to the symmetry of current. These two points before and after the commutation instant are offset by the same phase angle from the commutation point, and they are equidistant from the commutation point. This method of offsetting the sampling points by equidistant phase is a common technique in traditional correction methods based on current symmetry. However, when this shift technique is adopted to a motor with asymmetric back-EMF, as shown in Fig. 5(d), the currents at sampling points A and B are no longer equal, which will undoubtedly affect the correction accuracy of commutation deviation. Therefore, if the principle of current consistency before and after commutation is used to correct the commutation deviation of the asymmetric back-EMF motor, the freewheeling effect of the commutation process must be analyzed and eliminated.

C. Current Distortion in Commutation Process

Due to the existence of the winding inductance, the disappearing of the turn-OFF phase current and the establishment of the turn-ON phase current result in a short-time current distortion after commutation. Taking the commutation from AB phase conduction to ac phase conduction as an example, Fig. 6 shows the current path during the commutation process. The current in the turn-OFF phase (also known as freewheeling) is circulated through the antiparallel diode in the three-phase bridge without

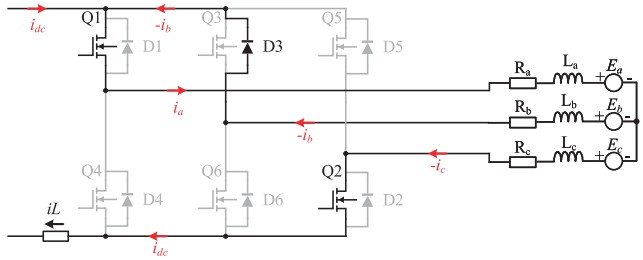


Fig. 6. Schematic diagram of current with freewheeling during commutation.

passing through the dc-link circuit. When the freewheeling occurs, $U_{ag} = U_{\rm dc}$, $U_{bg} = U_{\rm dc}$, $U_{cg} = 0$, combining with (1), the following equation can be obtained:

$$\begin{cases} \frac{di_a}{dt} + i_a \frac{R}{L} = \frac{U_{\rm dc} - 2E_a + E_b + E_c}{3L} \\ \frac{di_b}{dt} + i_b \frac{R}{L} = \frac{U_{\rm dc} - 2E_b + E_c + E_a}{3L} \\ \frac{di_c}{dt} + i_c \frac{R}{L} = \frac{-2U_{\rm dc} - 2E_c + E_a + E_b}{3L} \end{cases}. \tag{12}$$

The purpose here is to analyze the time of freewheeling current caused by the inductance and resistance in the commutation process, not the real transition time to the strictly current stable state, so the back-EMF can be considered as a constant value when solving the current. The expression of dc-link current in the commutation freewheeling process is

$$i_{\rm dc} = -i_c = \left(\frac{-2U_{\rm dc} - 2E_c + E_a + E_b}{3R} \right) e^{-\frac{R}{L}t} - \frac{-2U_{\rm dc} - 2E_c + E_a + E_b}{3R}. \tag{13}$$

Note that, for a single current sensor drive system, the detected dc-link current does not include the turn-OFF phase current, but only the turn-ON phase current.

When the freewheeling finishes, $U_{ag} = U_{\rm dc}, U_{cg} = 0, i_b = 0$, and $i_a = -i_c$, combining with (1), the steady current can be obtained as

$$i_{\rm dc_steady} = \frac{U_{\rm dc} - E_a + E_c}{2R}. \tag{14}$$

According to (13), the time $t_{\rm dc}$ when $i_{\rm dc}$ increases from 0 to $i_{\rm dc_steady}$ can be solved by making $i_{\rm dc} = i_{\rm dc_steady}$ and

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TABLE I
MSCMG MOTOR PARAMETERS

Parameter	Value
DC power supply	28 V
Moment of Inertia	0.13 N·m·s^2/rad
Phase back-EMF Coefficient	0.7744 V/k·RPM
Phase Resistance	0.5 Ω
Phase Inductance	33 μH
PWM Modulation Frequency	20 KHz
Pole Pairs	4

replacing $e^{-\frac{R}{L}t}$ with Taylor approximation $1-\frac{R}{L}t$ as

$$t_{\mathrm{dc}}=\frac{3L\left(U_{\mathrm{dc}}-E_a+E_c\right)}{2R\left(2U_{\mathrm{dc}}+2E_c-E_a-E_b\right)}. \tag{15}$$

When the commutation has just been completed, there are $E_a\approx E_b\approx -E_c$. $t_{\mathrm{dc}}\approx\frac{3L(U_{\mathrm{dc}}-2E_a)}{2R(2U_{\mathrm{dc}}-4E_a)}\leq\frac{3L}{4R}$ can be deduced within full-speed range. Considering the motor parameters in Table I, the commutation distortion time t_{dc} can be less than 0.05 ms.

IV. COMMUTATION DEVIATION CORRECTION METHOD

The proposed commutation deviation correction method in this article includes CR, OCPE, and phase shifter. The OCPE estimates the phase of the optimal commutation point deviating from ZCP according to the current value after commutation obtained by CR. Then, the phase shifter completes the correction of commutation deviation.

A. DC-Link Current Reconstructor

In order to eliminate the effect of current distortion in the commutation process, CR reconstructs the ideal current after commutation according to extrapolation of steady-state current after commutation. The theoretical basis of waveform extrapolation is the approximation of signal in a small range near the reconstruction point. When the speed is no more than 20 000 r/min, the phase angle of the motor commutation process is less than 0.42 rad, which is far less than 2π rad. So the current signal can be approximated by linear function according to Taylor expansion near the commutation point as

$$i=kt+b \tag{16}$$

where k is linear coefficient, b is linear bias, and t is the time from last commutation point. At this time, the current value at $t=0$ can be obtained by currents at only two sampling points in the steady state. As shown in Fig. 7, two red pentagons S_1 and S_2 represent the current sampling points at the steady state, red dotted lines represent the linear approximation function curves obtained from two current sampling points, red triangle S_0 represents the current values after commutation based on the linear extrapolation method, and blue triangle represents the current values before commutation. In order to use two current sampling points to more accurately approximate the linear tangent function of the current curve at the sampling point, the two sampling points should be as close as possible

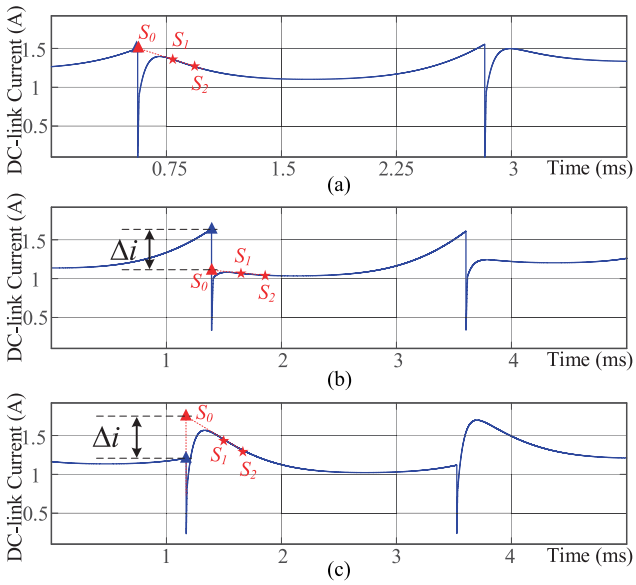


Fig. 7. Principle diagram of dc-link current reconstruction during commutation. (a) With optimal commutation. (b) With advanced commutation. (c) With lagged commutation.

while maintaining the current discrimination. Considering the sampling frequency of the current signal, the sampling interval is set as two sampling periods. Such two sampling points can guarantee the linear tangent approximate function of current waveform. The current sampling point control in this method is triggered by the microprocessor. After each commutation, the microprocessor will time and trigger the AD module to read the current sampling value according to the preselected current sampling position according to the real-time speed. Since the trigger control of the sampling point is homologous to the motor commutation control, the current must can be sampled at a preset position as long as the motor is commutated.

Linear parameters obtained from S_1 and S_2 current sampling points are as

$$\begin{cases} i_{S_1}=kt_{S_1}+b \\ i_{S_2}=kt_{S_2}+b \end{cases}\Rightarrow\begin{cases} k=\frac{i_{S_1}-i_{S_2}}{t_{S_1}-t_{S_2}} \\ b=i_{S_1}-\frac{i_{S_1}-i_{S_2}}{t_{S_1}-t_{S_2}}t_{S_1} \end{cases}. \tag{17}$$

In addition, the reconstructed current after commutation when $t=0$ is as

$$i_{\mathrm{after}}=kt_{=0}+b. \tag{18}$$

B. Optimal Commutation Phase Estimator

The OCPE estimates the phase angle between the optimal commutation point and the previous ZCP based on the principle of current consistency before and after commutation by a PI regulator. The integral function in the correction algorithm can eliminate the accidental error caused by current sampling to ensure the final correction effect of commutation deviation. The OCPE is structured as

$$\hat{\varphi}=\varphi_0+K_p\Delta i+K_i\int\Delta i \tag{19}$$

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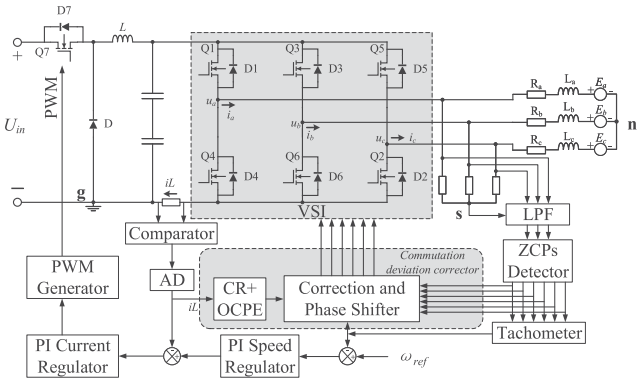


Fig. 8. Commutation deviation correction block diagram of BLDCM with asymmetric three-phase back-EMF.

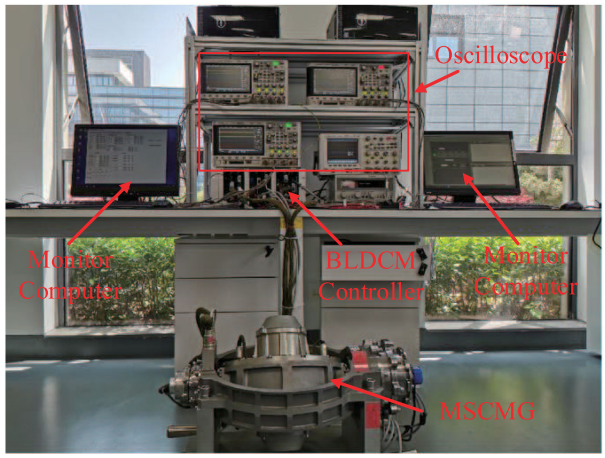


Fig. 9. MSCMG motor experiment system.

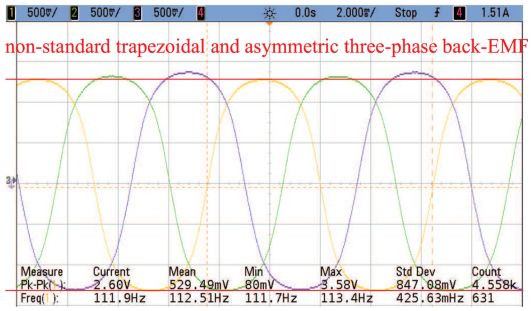


Fig. 10. Actual measured three-phase back-EMF.

where $\hat{\varphi}$ represents estimated phase angle, φ_0 equals to 30°, which is the initial phase shift between commutation point and ZCP, $\Delta i = i_{\text{before}} - i_{\text{after}}$ represents the difference between the measured current before commutation and reconstructed current after commutation, and K_p and K_i are the estimator coefficients, respectively.

C. Commutation Deviation Correction

According to the estimated phase angle between optimal commutation points and ZCPs, the phase shifter generates the commutation points, and the control system switches the voltage source inverter (VSI) according to the commutation table to drive the motor. Fig. 8 shows the overall commutation deviation correction block diagram of BLDCM with asymmetric three-phase back-EMF. Due to the asymmetric characteristics of three-phase back-EMF, the deviations of six commutation points are different. Correspondingly, each commutation point needs to be equipped with an independent correction system.

V. Experiments

A. Setup

Fig. 9 illustrates the configuration of the MSCMG system for experiments. The BLDCM is built inside the gyro cabin, and its control system is integrated in the controller cabinet. Detailed parameters of the BLDCM for MSCMG are listed in Table I.

B. Actual Measured Back-EMF Verification

The actual measured back-EMF waveforms for the motor used in MSCMG are shown in Fig. 10. It can be seen that the three-phase back-EMF is asymmetrical. By offline measurement, the back-EMF parameters as in (2) can be obtained, and the detailed results are listed in Table II.

C. CR and OCPE Experiment Results

This experiment evaluates the experimental effect of the CR designed in this article. As mentioned in the article, CR is actually an extrapolation processing method of signal. Fig. 11

TABLE II
Detailed Measured Back-EMF Parameters

A_{a1}	A_{b1}	A_{c1}	A_{a3}	A_{b3}	A_{c3}
1	≈ 0.9	≈ 1	≈ 0.21	≈ 0.19	≈ 0.21
A_{a5}	A_{b5}	A_{c5}	D_a	D_b	D_c
≈ 0.039	≈ 0.035	≈ 0.038	≈ 0	≈ 0	≈ 0.05

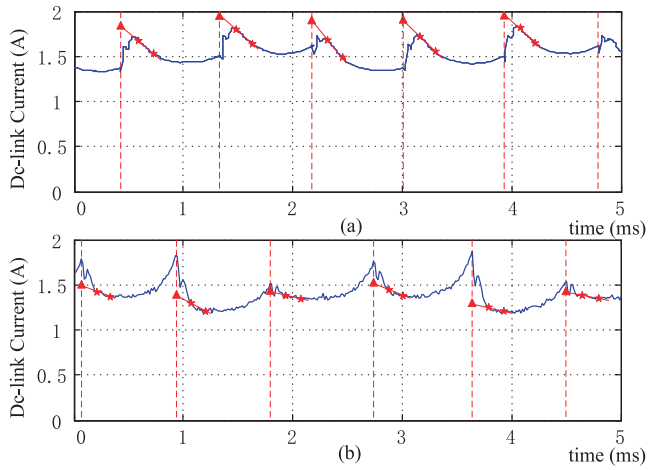


Fig. 11. Reconstructed currents in experiments based on the proposed CR. (a) With advanced commutation. (b) With lagged commutation.

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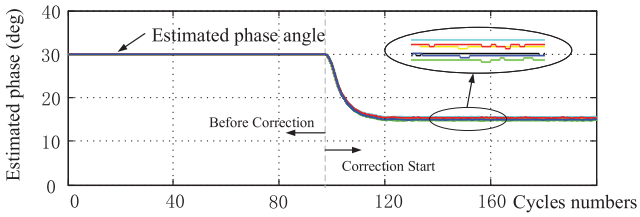


Fig. 12. Estimated optimal phases of six commutation points in experiments based on the proposed OCPE.

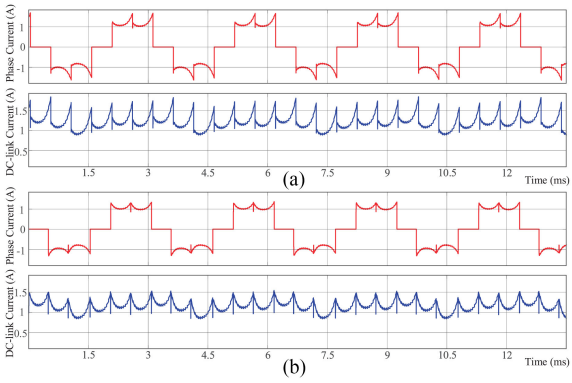


Fig. 13. Phase current and dc-link current at 5000 r/min. (a) Without correction. (b) With proposed correction.

shows the original current signal, two sampling points for reconstruction, and the reconstructed current value after commutation on a single figure. The blue line in the figure represents the original current signal, the two pentagons represent the sampling points used for reconstruction, and the triangle represents the reconstructed current value. The red short line is a three-point line, which is a linear extrapolation function. It can be seen that the reconstructed current at almost every commutation point is different due to the asymmetry of three-phase back-EMF.

Based on the difference between the reconstructed current after commutation and the measured current before commutation, the OCPE uses (17) to force the current difference to zero and estimate the optimal commutation phase shift of six commutation points in each cycle according to the principle of current consistency. Fig. 12 shows the process of estimator convergence from the initial state to the estimated value. By choosing reasonable parameters, the whole estimated value can be stabilized in 30–40 electrical cycles. The difference of steady values shows the phase difference among six commutation points caused by asymmetrical three-phase back-EMF. This is consistent with the slight deviation of ZCPs shown in Fig. 3.

D. Commutation Deviation Correction Experiment Results

Fig. 13 shows the comparison of phase current and dc-link current before and after correction under 5000 r/min. It can be clearly seen that the current waveform before correction has obvious distortion, the peak value of current has increased significantly, and the pulsation of dc-link current has also increased

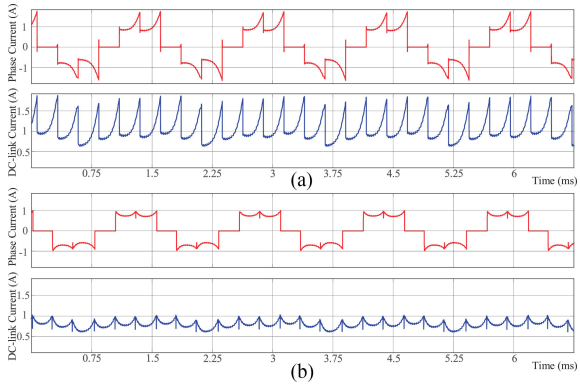


Fig. 14. Phase current and dc-link current at 10 000 r/min. (a) Without correction. (b) With proposed correction.

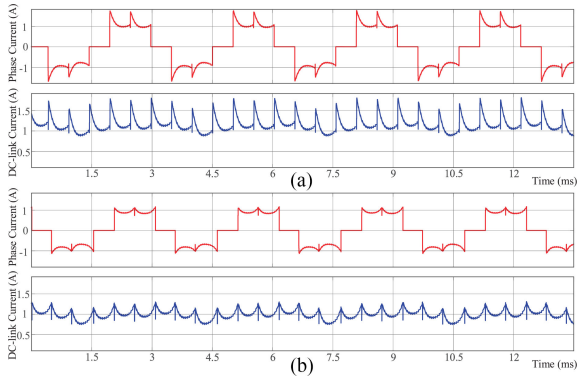


Fig. 15. Phase current and dc-link current when advanced 10° commutation. (a) Without correction. (b) With proposed correction.

significantly. There is no doubt that this will bring vibration and noise to the motor. The corrected current restores to the ideal state and only asymmetric back-EMF affects the waveform. Furthermore, the experiment results under 10 000 r/min are shown in Fig. 14. Compared with Fig. 13, the phase current under high speed has more obvious distortion, and even freewheeling currents in nonconducting phase occur, which will lead to negative torque and reduce motor efficiency. After the application of the proposed correction method, the current has been significantly improved and restored to an ideal state. In addition, in order to demonstrate the phase adjustment capability of the proposed method, a phase advance correction experiment is set artificially in Fig. 15. The results show that the current waveform of the phase advance commutation is opposite to that of the phase lag commutation, and the distortion is obvious, which affects the efficiency of the motor. After applying the proposed method, the phenomenon of advanced commutation is eliminated and the waveform is restored to the ideal state.

E. Experimental Results Compared With Traditional Correction Methods

A comparative experiment based on the traditional symmetry correction method and the proposed method was also carried out. Fig. 16 shows the comparison results. It can be seen from

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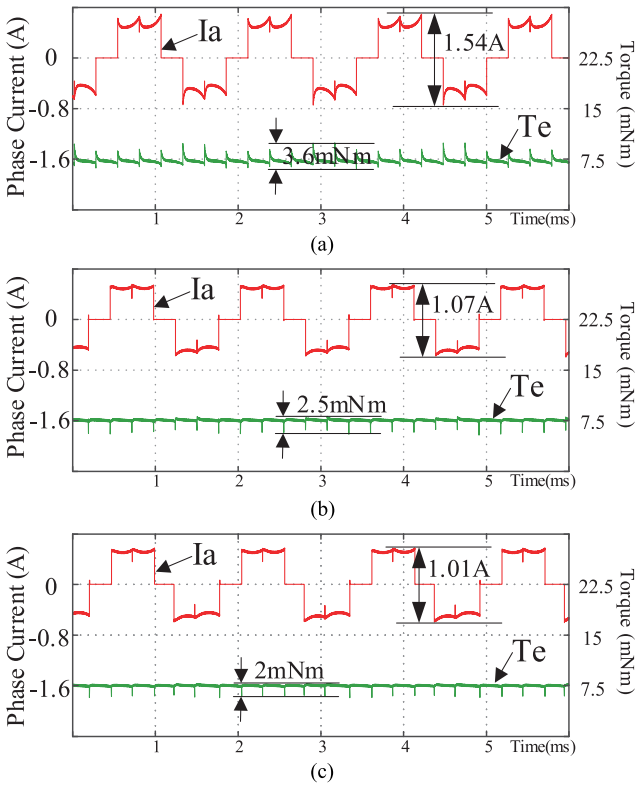


Fig. 16. Comparison of phase current and torque in different methods. (a) Without correction. (b) With traditional symmetry-based correction. (c) With proposed correction.

the figure that when the conventional and proposed correction methods are applied, the peak-to-peak value of the phase current is significantly reduced compared with the uncorrected, and the pulsation at the commutation of the torque is also significantly reduced. Compared with the traditional method, the phase current and torque ripple peak-to-peak value of the proposed method are further reduced slightly, which is consistent with the small commutation deviation caused by the asymmetric back-EMF that can be eliminated by the proposed method.

VI. Conclusion

Aiming at sensorless commutation deviation of BLDCM with asymmetric three-phase back-EMF, a commutation deviation correction method based current consistency was proposed in this article. First, this article analyzed the influencing factors of commutation deviation under asymmetric three-phase back-EMF, and pointed out the shortcomings caused by commutation deviation. On this basis, a correcting commutation deviation method was proposed. At the same time, in order to eliminate the influence of waveform distortion caused by winding inductance during commutation, a dc-link CR in the commutation process was proposed. Based on the reconstructed current value after commutation, an optimal OCPE based on commutation current consistency was designed to estimate the position of each optimal commutation point. The main contribution of the proposed correction method was that only single current sensor was used to correct the sensorless commutation deviation of motor with three-phase asymmetric back-EMF for the first time. It should be noted that, precisely because of the considering of asymmetry, the proposed method abandons the traditional method based on current or voltage symmetry, which does not consider asymmetry and cannot eliminate the deviation caused by asymmetry. The proposed method has universal adaptability, regardless of whether the back-EMF was symmetric or asymmetric. Finally, experiment results confirmed that the method can achieve high-precision sensorless commutation deviation correction at each commutation point in the case of asymmetric three-phase back-EMF.

References

[1] M. Z. Youssef, "Design and performance of a cost-effective BLDC drive for water pump application," *IEEE Trans. Ind. Electron.*, vol. 62, no. 5, pp. 3277–3284, May 2015.
[2] J. Fang and Y. Ren, "High precision control for a single gimbal magnetically suspended control moment gyro based on inverse system method," *IEEE Trans. Ind. Electron.*, vol. 58, no. 9, pp. 4331–4342, Sep. 2011.
[3] P. Damodharan and K. Vasudevan, "Sensorless brushless dc motor drive based on the zero-crossing detection of back electromotive force (EMF) from the line voltage difference," *IEEE Trans. Energy Convers.*, vol. 25, no. 3, pp. 661–668, Sep. 2010.
[4] G. Liu, C. Cui, K. Wang, B. Han, and S. Zheng, "Sensorless control for high-speed brushless dc motor based on the line-to-line back-EMF," *IEEE Trans. Power Electron.*, vol. 31, no. 7, pp. 4669–4683, Jul. 2016.
[5] J. X. Shen and S. Iwasaki, "Sensorless control of ultrahigh-speed PM brushless motor using PLL and third harmonic back-EMF," *IEEE Trans. Ind. Electron.*, vol. 53, no. 2, pp. 421–428, Apr. 2006.
[6] J. M. Liu and Z. Q. Zhu, "Improved sensorless control of permanent magnet synchronous machine based on third-harmonic back-EMF," *IEEE Trans. Ind. Appl.*, vol. 50, no. 3, pp. 1180–1187, May/Jun. 2013.
[7] P. B. Beccue, S. D. Pekarek, B. J. Deken, and A. C. Koenig, "Compensation for asymmetries and misalignment in a hall-effect position observer used in PMSM torque-ripple control," *IEEE Trans. Ind. Appl.*, vol. 43, no. 2, pp. 560–570, Mar.–Apr. 2007.
[8] N. Samoylenko, Q. Han, and J. Jatskevich, "Dynamic performance of brushless DC motors with unbalanced hall sensors," *IEEE Trans. Energy Convers.*, vol. 23, no. 3, pp. 752–763, Sep. 2008.
[9] J. S. Park, J. H. Choi, and J. Lee, "Compensation method of position signal error with misaligned hall-effect sensors of BLDC motor," *J. Elect. Eng. Technol.*, vol. 11, no. 4, pp. 889–897, Jul. 2016.
[10] J. S. Park and K.-D. Lee, "Online advanced angle adjustment method for sinusoidal BLDC motors with misaligned hall sensors," *IEEE Trans. Power Electron.*, vol. 32, no. 11, pp. 8247–8253, Nov. 2017.
[11] J. X. Shen and K. J. Tseng, "Analyses and compensation of rotor position detection error in sensorless PM brushless DC motor drives," *IEEE Trans. Energy Convers.*, vol. 18, no. 1, pp. 87–93, Mar. 2003.
[12] Q. Jiang, C. Bi, and R. Huang, "A new phase-delay-free method to detect back-EMF zero-crossing points for sensorless control of spindle motors," *IEEE Trans. Magn.*, vol. 41, no. 7, pp. 2287–2294, Jul. 2005.
[13] G. H. Jang and M. G. Kim, "Optimal commutation of a BLDC motor by utilizing the symmetric terminal voltage," *IEEE Trans. Magn.*, vol. 42, no. 10, pp. 3473–3475, Oct. 2006.
[14] X. Wu, B. Zhou, F. Song, F. Chen, and J. Wei, "A closed loop control method to correct position phase for sensorless brushless DC motor," in *Proc. Int. Conf. Elect. Mach. Syst.*, Oct. 17–20, 2008, pp. 1460–1464.
[15] Y. Wu, Z. Deng, X. Wang, X. Ling, and X. Cao, "Position sensorless control based on coordinate transformation for brushless DC motor drives," *IEEE Trans. Power Electron.*, vol. 25, no. 9, pp. 2365–2371, Sep. 2010.
[16] W. Li, J. Fang, H. Li, and J. Tang, "Position sensorless control without phase shifter for high-speed BLDC motors with low inductance and non-ideal back-EMF," *IEEE Trans. Power Electron.*, vol. 31, no. 2, pp. 1354–1366, Feb. 2016.
[17] M. Lee and K. Kong, "Fourier-series-based phase delay compensation of brushless DC motor systems," *IEEE Trans. Power Electron.*, vol. 33, no. 1, pp. 525–534, Jan. 2018.

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[18] C. Gu, X. Wang, X. Shi, and Z. Deng, "A PLL based novel commutation correction strategy for high-speed brushless DC motor sensorless drive system," *IEEE Trans. Ind. Electron.*, vol. 65, no. 5, pp. 3752–3762, May 2018.
[19] X. Zhou, Y. Zhou, C. Peng, F. Zeng, and X. Song, "Sensorless BLDC motor commutation point detection and phase deviation correction method," *IEEE Trans. Power Electron.*, vol. 34, no. 6, pp. 5880–5892, Jun. 2019.
[20] H. T. Li, X. Ning, and W. Z. Li, "Implementation of a MFAC based position sensorless drive for high speed BLDC motors with nonideal back-EMF," *ISA Trans.*, vol. 67, pp. 348–355, Mar. 2017.
[21] J. S. Park, K.-D. Lee, S. G. Lee, and W.-H. Kim, "Unbalanced ZCP compensation method for position sensorless BLDC motor," *IEEE Trans. Power Electron.*, vol. 34, no. 4, pp. 3020–3024, Sep. 2019.
[22] H. C. Chen and H. H. Huang, "Design of buck-type current source inverter fed brushless DC motor drive and its application to position sensorless control with square-wave current," *IET Elect. Power Appl.*, vol. 7, no. 5, pp. 416–426, Jul. 2013.



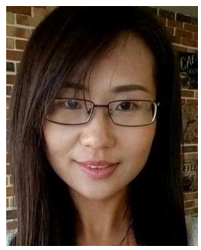
Gang Liu received the B.S. and M.S. degrees in industrial automation from Shandong University, Jinan, China, in 1992 and 1998, respectively, and the Ph.D. degree in water conservancy and hydropower engineering from the Dalian University of Technology, Dalian, China, in 2001.

He is currently a Ph.D. Supervisor with the School of Instrumentation Science and Optoelectronics Engineering, Beihang University (BUAA), Beijing, China. His research interests include spacecraft attitude control, high-speed magnetically suspended motor, and ultrahigh sensitivity atomic magnetic field measurement.



Xi Chen received the B.S. degree in automation from Shandong University, Jinan, China, in 2014. He is currently working toward the Ph.D. degree in precision instruments and machinery with the School of Instrumentation Science and Optoelectronics Engineering, Beihang University (BUAA), Beijing, China.

His research interests include the spacecraft attitude actuator control and motor control.



Xinxiu Zhou (M'14) received the B.S. degree from Yanshan University, Hebei, China, in 2006, and the Ph.D. degree from Beihang University (BUAA), Beijing, China, in 2013, both in torque control of flywheel motor.

She is currently a Research Member with the Key Laboratory of Fundamental Science for National Defense, Novel Inertial Instrument and Navigation System Technology, Beijing, China. Her research interests include the motor control, power electronics, and spacecraft attitude control.



Shiqiang Zheng (M'12) was born in Shandong, China, in 1981. He received the Ph.D. degree in electrical and electronics engineering from the Beijing University of Aeronautics and Astronautics (now Beihang University), Beijing, China, in 2011.

He is currently an Associate Professor of Magnetic Bearing Control Technology with the School of Instrumentation and Optoelectronic Engineering, Beihang University, where he is also with the Beijing Engineering Research Center of High-Speed Magnetically Suspended Motor Technology and Application. From 2015 to 2016, he was a Visiting Scholar with the Advanced Robotics Center, National University of Singapore, Singapore. His research interests include magnetic bearing technology in high-speed rotating machinery and attitude control actuators of spacecraft.

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